

Project Reader

Write a program that will read-in lisp expressions generated by the Lisp grammar and store them in a character array.

Make sure you take care of the following details:

[

Just follow instructions (1) to (7);
if not you will regret later

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1) When you start your Program, issue a prompt "->"

2) It should read input as characters, one at a time
OR
you may choose to read in the whole line at once.

In either case the all the input **MUST be stored in a character array**, 5000 long, or a string.

3) Skip all the characters until end of line (comments) when a ";" (semicolon) is encountered.

eg: (define f(x y) ; this function add two numbers
(+ x y))

Here you start with a prompt "->" when the first line
"(define f(x y) ; this function add two numbers"
is entered you will only store
"(define f(x y)"

and throw away the rest: "this function add two numbers"

Now, since the parenthesis count does not match issue the prompt ">" (see (4)).

4) The input can span several lines, and if ever the program sees an eoln (end of line character) and there is more "(" than ")" you should issue a the prompt ">".

The current input is completed when the "(" count matches the ")" parenthesis count. At that point you issue a "-->" prompt.

However, the program should not terminate. (see (7))

i.e. in the above example when you input the second line

"(+ x y))"
the current input is completed and the prompt "-->" is issued.

BEFORE YOU READ THE NEXT INPUT MAKE SURE EVERYTHING IS RESET, i.e. clear the character array etc.

- 5) Make sure you remember the length of the input (for error checking)
- 6) Add a special character to end of the input in the character array (#)
Again for error checking.
- 7) The program should loop until it sees a "quit"
at this point your program terminates.

Come to class for more details